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ABSTRACT

Understanding the influence of a crowded intracellular environment on the structure and solvation of DNA functionalized gold nanoparticles (ss-DNA AuNP) is necessary for designing applications in nanomedicine. In this study, the effect of single (Gly, Ser, Lys) and mixture of amino acids (Gly+Ser, Gly+Lys, Ser+Lys) at crowded concentrations is examined on the structure of the ss-DNA AuNP using molecular dynamics simulations. Using the structural estimators such as pair correlation functions and ligand shell positional fluctuations, the solvation entropy is estimated. Combining the AuNP–solvent interaction energy with the solvation entropy estimates, the free energy of solvation of the AuNP in crowded solutions is computed. The solvation entropy favours the solvation free energy which becomes more favourable for larger effective size of AuNP in crowded solutions relative to that in water. The effective size of AuNP depends on the different propensity of the crowders to adsorb on Au surface, with the smallest crowder (Gly) having the highest propensity inducing the least effective AuNP size as compared to other single crowder solutions. In mixed crowded solutions of amino acids of variable size and chemistry, distinctive local adsorption of the crowders on the gold surface is observed that controls the additive or non-additive crowding effects which govern an increase (in Gly+Ser) or decrease (in Gly+Lys) in nanoparticle effective size respectively. The results shed light into the fundamental understanding of the influence of intracellular crowding on structure of ss-DNA AuNP and plausible employability of crowding as a tool to design programmable self-assembly of functionalized nanoparticles.

I. INTRODUCTION

Tailoring the properties of bio-functionalized metal nanoparticles, such as ss-DNA capped gold nanoparticle (ss-DNA AuNP),[1,2] requires an understanding of their behaviour in biological fluids to design applications in biosensing[3–5] and nanomedicine.[6–8] This involves accounting for the crowding effects inside a living cell and obtaining a fundamental understanding of the structure and function of these nanoparticles inside the crowded milieu. The cellular interior is a tightly packed environment of large-sized macromolecules such as proteins, lipids, sugars that occupy up to 40% of the cellular volume, reducing the total available free volume for bioprocesses. Such an environment is expected to induce compaction or self-assembly in biomolecules (or nanoparticles) since that would reduce the solvent excluded volume (steric) effects[9,10] of the nanoparticle, rendering larger free configurational volume to the crowders making it entropically favourable.[11–13] However, an interplay of steric and non-specific interactions of the crowders with the nanoparticle (or biomolecule) can play a crucial role in determining the structure of the nanoparticle, that needs to be further explored.[14–19] Therefore, an interplay of such size (entropic) and soft interaction (energetic) effects of the crowded environment on the nanoparticle structure needs to be investigated. Prior studies have used structural estimators of entropy based on pair correlation functions to evaluate the entropy of solvation of such nanoparticles that could be one useful approach to attain molecular insights.[20]

The biomedical applications involving self-assembly of ss-DNA AuNPs can be induced either by using a linker DNA that hybridizes with complementary linker strand on the other AuNP[21,22] or via non-base pairing interactions between non-complementary DNA strands in the absence of the linker DNA.[23] One of the experimental studies showed that by increasing the ionic strength and concentration of divalent cations in the solution, the increasing surface tension of AuNPs could be diminished and non-base pairing in DNA could drive the self-assembly of the AuNPs.[24] The influence of a crowding environment in inducing AuNP aggregation via non-base pairing interactions between the DNA grafting AuNPs has been recently examined. It was shown that polyethylene glycol (PEG) as the crowders can induce aggregation of AuNPs in the absence of the linking DNA strands, which could be tuned by modulating the ionic strength in the solution.[25] In another study it was found that PEG (used as crowder) increased the melting point of aggregated AuNPs in the absence of linker DNA and a broad transition was observed.[26] This broad melting transition was attributed to the depletion attraction forces that arise due to the presence of large-sized crowders in these solutions that push the AuNP together. These studies imply the crucial role of the crowded environment in modulating self-assembly of ss-DNA AuNPs. However, the influence of *mixed crowded* environment on the structural features and solvation thermodynamics of the ss-DNA AuNP remains elusive. Moreover, simulation studies have focused on examining the structure and thermodynamics of DNA-capped nanoparticles via atomistic[21,27] and coarse-grained simulations,[28,29] the role of crowding environment has not been explored fully using simulations.

In this study, we examine the effect of crowded concentrations of amino acids, glycine (Gly), serine (Ser), lysine (Lys) and binary mixtures of these amino acids (Gly+Ser, Gly+Lys, Ser+Lys), on the structural changes and solvation thermodynamics of the AuNP capped with ss-DNA. In a previous study, the melting transition of DNA-capped AuNPs was found to be not fully dependent on the molecular weight of crowder, implying the possibility of small sized crowders to induce the same effect.[25] We ask, (1) how the size and chemical nature of crowders influence the ligand shell structure and solvation thermodynamics?, (2) is the effect of a mixed crowded environment different from that of single crowder environment? We perform molecular dynamics simulations of these solutions and investigate the conformational preferences of the ss-DNA strands (composed of ten thymines) on AuNP surface. The solvation entropy is computed using the pair correlation function for the nanoparticle and solvent (crowders and water) when the solvent reoroganizes around the solute. The ligand shell entropy is computed using the covariance in the positions of DNA atoms. Using the solvation entropy and energetic interactions between the nanoparticle and solvent (crowders and water), the free energy of solvation of the nanoparticle is investigated in the crowded solutions.

II. COMPUTATIONAL DETAILS

A. System setup

The system investigated in this study comprises of a gold nanoparticle functionalized with four ss-DNA chains in water and in aqueous solutions of crowders modelled by Glycine (Gly) or Serine (Ser) or a mixture of both the crowders at high packing fractions of up to 18%–24%. The gold nanoparticle (AuNP) is composed of 201 gold atoms and has a truncated octahedral geometry with eight (111) surfaces and six (100) surfaces. The (111) surface is functionalized with four ssDNA strands, each composed of ten nucleotides i.e. ten thymines (T_{10}), using a six-carbon alkylthiolate linker [$-S(CH_2)_6-$]. The canonical B-DNA structure is used as a model for the initial helical parameters of the ss-DNAs. The all-atom CHARMM27 force field[30] parameters were used for modeling the ss-DNA and water was modelled using modified TIP3P potential. Amino acids, ss-DNA and ions were modelled using the CHARMM27 parameters. Additionally, NBFIX reparameterization was used,[31,32] to circumvent any artificial aggregation in the solution. The reparameterization involved incrementing the Lennard-Jones size parameter by 0.08 Å for the interaction between N of amine and carboxyl O of the zwitterionic termini of the amino acids. The overall charge of the system was neutralized by adding 40 Na^+ ions. In addition to that, 0.5M concentration of NaCl is added to the system. For solutions consisting of Lys (positively charged side chain), additional 2000 Cl^- were added to maintain the charge neutrality of the solution. Lennard-Jones potentials are used to describe the interactions between gold atoms, with the parameters σ (0.2620 nm) and ϵ (22.13 kJ mol^{-1}) obtained from the literature.[33] The parameters for the alkylthiol are obtained from the Ref. 33 The details of the systems simulated in this work are summarized in Table I. Representative snapshots of the system are shown in Figs. 1 and S1.

B. Simulation details

Molecular dynamics (MD) simulations were performed for the systems using GROMACS 2021.4.[34] As the first step, the energy of the system was minimized using the steepest descent algorithm. The system was then subjected to equilibration in the canonical (NVT) ensemble for 1 ns. The system was coupled to a velocity rescaling thermostat with a coupling constant of 0.1 ps to maintain the temperature at 300 K. The positions of the gold atoms were restrained using a harmonic potential with a force constant of 1000 kJ mol^{-1} nm^{-1}. An additional equilibration was carried out in isothermal–isobaric ensemble (NPT) for 1 ns. Langevin dynamics algorithm was used with a damping coefficient of 1 ps^{-1} to control the temperature at 300 K. The pressure was controlled by Parrinello–Rahman barostat at 1 atm with a coupling constant of 0.2 ps. After equilibration, a production run was carried out in the NPT ensemble for 200 ns using Langevin dynamics. The time step was set to 2 fs, and the coordinates were saved every 10 ps. The long-range electrostatic interactions were treated using the Particle Mesh Ewald (PME) method[35] with a grid size of 0.12 nm. The cutoff distance for the real-space electrostatics and van der Waals interactions was 1.2 nm. The energies of interaction between different

TABLE I. Molecular dynamics simulations system setup. All systems have been simulated in a crowded concentration with packing fraction between ~17% to ~34%. The average box size is obtained from NPT production run.

System	No. of water molecules	No. of crowders	Box size (nm)	Packing fraction
Water	31 503	⋯	9.827	⋯
Gly	23 420	2000	9.468	17.5%
Ser	20 835	2000	9.462	24.0%
Lys	12 732	2000	9.441	33.87%
Gly+Ser	22 127	1000 + 1000	9.459	20.7%
Gly+Lys	18 017	1000 + 1000	9.476	29.81%
Ser+Lys	16 727	1000 + 1000	9.477	31.5%

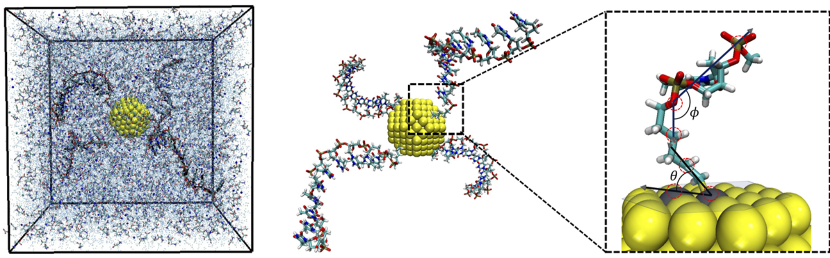

FIG. 1. Representative snapshot of the box of ss-DNA capped AuNP system in crowded solution of Glycine used for simulations. The AuNP is shown capped with four ss-DNA T_{10} chains. The magnified snapshot shows the orientation angles (θ and ϕ) of the ss-DNA chain on the Au surface that have been computed in this work. The position vectors for computing the angles are defined by the positions of atoms circled with red color (see Observables section).

components such as gold–crowders, gold-water, ligand–water and ligand–crowder were computed using reaction-field method using dielectric constant for modified TIP3P set as 102.4. The SHAKE method was used to hold rigid covalent bonds containing hydrogen. The last 100 ns of the production run is considered for the analysis. Error bars were computed by dividing the trajectories into blocks of equal size and standard deviation was computed for the mean value of the blocks from the overall average value of the observable.

III. THEORY

The solvation free energy of a structureless solute can be described by the reversible work done to transfer the solute from vacuum to a fixed point in solution.[36] According to the statistical mechanics formulation, the standard state would be to fix the location of the solute in the solvent at constant temperature and volume, to eliminate the translational contribution of the solute to the total solvation entropy. The solvation energy (ΔU_{solv}) and entropy (ΔS_{solv}) would then arise due to the solvent reorganization around the solute. The free energy of solvation can be defined as,

$$\Delta G_{solv} = \Delta U_{solv} - T\Delta S_{solv} \tag{1}$$

The $P\Delta V$ is negligible and can be neglected. In our work, the gold nanoparticle is rigid and its coordinates are fixed in the solvent, implying that the solute has no translational or rotational degrees of freedom. The ss-DNA chains are allowed to fluctuate and reorganize under the effect of the solvent. The solvation energy is given by the sum of the interaction energies of the nanoparticle with the solvent (water and crowders). This can be written as sum of contributions arising from the interactions of solute–water (ΔU_{nw}) and solute–crowder (ΔU_{nc}) which can be further written as sum of contributions from interactions of gold–water (U_{g-w}), gold–crowder (U_{g-c}), ligand–water (U_{l-w}), ligand–crowder (U_{l-c}) and the intra-chain and inter-chain energy of the ss-DNA (U_{l-l}).

$$\begin{aligned}\Delta U_{solv} &= \Delta U_{nw} + \Delta U_{nc}\\ &= U_{g-w} + U_{g-c} + U_{l-w} + U_{l-c} + U_{l-l}\end{aligned} \tag{2}$$

Here, the interaction with water includes the interactions of gold or the ligand with the ions as well i.e. $U_{g-w} = U_{g-wat} + U_{g-ions}$ and $U_{l-w} = U_{l-wat} + U_{l-ions}$. Since the gold core positions do not significantly fluctuate during the solvation, we need not consider the intra-core energy in the above expressions.

The contributions to the solvation entropy would arise due to solvent reorganization around the solute (ΔS^{reorg}_{solv}) and due to the conformational fluctuations in the ss-DNA chain (ΔS_L).

$$\Delta S_{solv} = \Delta S^{reorg}_{solv} + \Delta S_L \tag{3}$$

Here, the nanoparticle is rigid and no significant structural reorganization of the inorganic core is expected. Therefore, the internal structure of the nanoparticle can be neglected and the contribution due to the solute to entropy primarily comes from the conformational entropy of the ss-DNA chains given by, $\Delta S_L = S_L(\rho) - S_L(\rho = 0)$. The S_L can be determined using Schlitter's entropy formula described in Sec. III A. The solvent reorganization entropy, ΔS_{reorg}, can be defined as follows using the approach followed by Lazaridis.[37,38] Consider N_n solute (functionalized nanoparticle) solvated in N_w water molecules and N_c crowder molecules in solution. The ideal contribution to the ensemble-invariant entropy is given by,[39,40]

$$\begin{aligned}S_{id} &= S_{id,n} + S_{id,c} + S_{id,w}\\ S_{id} &= N_n\left(\frac{5}{2}k_B - k_B\ \ln\left(\rho_n\Lambda_n^3\right)\right) + N_c\left(\frac{5}{2}k_B - k_B\ \ln\left(\rho_c\Lambda_c^3\right)\right)\\ &\quad + N_w\left(\frac{5}{2}k_B - k_B\ \ln\left(\rho_w\Lambda_w^3\right)\right)\end{aligned} \tag{4}$$

where, ρ_n, ρ_c, ρ_w are respectively the number densities of the nanoparticle, crowders and water. The Λ_n, Λ_c and Λ_w are thermal de Broglie's wavelengths of the solute, crowders and water respectively. The excess entropy contribution is dominated by the pair entropy, where the three-body and higher-body terms in the multiparticle correlation expansion of the entropy can be neglected.

$$\begin{aligned}S_{ex} = &-k_BN_n\rho_w\int(g_{nw}\ \ln g_{nw} - g_{nw} + 1)d\mathbf{r}\\ &- k_BN_n\rho_c\int(g_{nc}\ \ln g_{nc} - g_{nc} + 1)d\mathbf{r}\\ &- k_BN_w\rho_w\int(g_{ww}\ \ln g_{ww} - g_{ww} + 1)d\mathbf{r}\\ &- k_BN_c\rho_c\int(g_{cc}\ \ln g_{cc} - g_{cc} + 1)d\mathbf{r}\\ &- k_BN_c\rho_w\int(g_{cw}\ \ln g_{cw} - g_{cw} + 1)d\mathbf{r}\\ &- k_BN_n\rho_n\int(g_{nn}\ \ln g_{nn} - g_{nn} + 1)d\mathbf{r}\end{aligned} \tag{5}$$

where, g_{nw}, g_{nc}, g_{ww}, g_{cc}, g_{cw}, g_{nn} are the radial distribution functions (rdfs) for the nanoparticle–water, nanoparticle–crowder, water–water, crowder–crowder, crowder–water and nanoparticle–nanoparticle interactions respectively. It is to be noted that the rdfs involving water include the ions as well. Therefore, with $S = S_{id} + S_{ex}$, the entropy of solvation can be obtained by taking the $\left(\frac{\partial S}{\partial N_n}\right)_{T,P}$ at infinite dilution limit where $N_n \to 1$ and $\rho_n \to 0$. For this we consider taking the derivatives of the terms in Eq. (4) and the first two terms of Eq. (5). The last term in Eq. (5) will not be relevant at infinite dilution of the solute. The third, fourth and fifth terms of Eq. (5) correspond to the solvent reorganization in bulk and will not be considered here. The partial molar volume of the solute at infinite dilution can be defined as,

$$\bar{v}_n^{\infty} = \left(\frac{\partial V}{\partial N_n}\right)_{T,P,\rho_n \to 0} \tag{6}$$

We obtain the $\left(\frac{\partial S}{\partial N_n}\right)_{T,P} = \bar{S}_n^{\infty}$ as,

$$\begin{aligned}\bar{S}_n^{\infty} &= \frac{3}{2}k_B - k_B \ln\left(\rho_n \Lambda_n^3\right) + k_B \bar{v}_n^{\infty}\left[\rho_n + \rho_c + \rho_w\right] - k_B \rho_w \\ &\times \int (g_{nw} ln g_{nw} - g_{nw} + 1) d\mathbf{r} - k_B \rho_c \int (g_{nc} \ln g_{nc} - g_{nc} + 1) \\ &\times d\mathbf{r} + k_B \bar{v}_n^{\infty} \rho_n \Big[\rho_w \int (g_{nw} \ln g_{nw} - g_{nw} + 1) d\mathbf{r} \\ &+ \rho_c \int (g_{nc} \ln g_{nc} - g_{nc} + 1) d\mathbf{r}\Big]\end{aligned} \tag{7}$$

At infinite dilution, $\rho_n \to 0$,

$$\begin{aligned}\bar{S}_n^{\infty} &= \frac{3}{2}k_B - k_B \ln\left(\rho_n \Lambda_n^3\right) + k_B \bar{v}_n^{\infty}\left[\rho_c + \rho_w\right] \\ &- k_B \rho_w \int (g_{nw} \ln g_{nw} - g_{nw} + 1) d\mathbf{r} \\ &- k_B \rho_c \int (g_{nc} \ln g_{nc} - g_{nc} + 1) d\mathbf{r}\end{aligned} \tag{8}$$

Using Kirkwood–Buff (KB) theory,[41] the $\bar{v}_n^{\infty}$ can be defined for ternary mixtures in terms of KB integrals of nanoparticle–water (G_{nw}) and nanoparticle–crowder (G_{nc}). The solution has another component of ions due to the salt. The interaction of the nanoparticle with the ions is included in the rdf g_{nw} or KB integral G_{nw} as also mentioned before. This is because the KB theory for tertiary solution is not trivial and would require rigourous theoretical framework and this forms part of our future study.

$$\bar{v}_n^{\infty} = k_B T \kappa_T - \rho_w \bar{V}_w G_{nw} - \rho_c \bar{V}_c G_{nc} \tag{9}$$

where, κ_T is the isothermal compressibility of the solution and $\bar{V}_w$ and $\bar{V}_c$ are respectively the partial molar volumes of water and crowders defined as below.[42]

$$\begin{aligned}G_{nw} &= 4\pi \int_0^R (g_{nw} - 1) r^2 dr \\ G_{nc} &= 4\pi \int_0^R (g_{nc} - 1) r^2 dr \\ \bar{V}_c &= \frac{1 + \rho_w (G_{ww} - G_{cw})}{\rho_w + \rho_c + \rho_w \rho_c (G_{ww} - 2G_{cw} + G_{cc})} \\ \bar{V}_w &= \frac{1 + \rho_c (G_{cc} - G_{cw})}{\rho_w + \rho_c + \rho_w \rho_c (G_{ww} - 2G_{cw} + G_{cc})}\end{aligned} \tag{10}$$

Substituting Eq. (9) in Eq. (8) we get,

$$\begin{aligned}\bar{S}_n^{\infty} &= \frac{3}{2}k_B - k_B \ln\left(\rho_n \Lambda_n^3\right) + k_B\left[\rho_c + \rho_w\right] \\ &\times \left(k_B T \kappa_T - \rho_w \bar{V}_w G_{nw} - \rho_c \bar{V}_c G_{nc}\right) \\ &- k_B \rho_w \int (g_{nw} \ln g_{nw} - g_{nw} + 1) d\mathbf{r} \\ &- k_B \rho_c \int (g_{nc} \ln g_{nc} - g_{nc} + 1) d\mathbf{r}\end{aligned} \tag{11}$$

The first term is due to the translational entropy of the solute and can be neglected. The term with κ_T is for the bulk solution and need not be considered here. Therefore, the solvent reorganization entropy at infinite dilution, ΔS_{solv}^{reorg} can be written as,

$$\begin{aligned}\Delta S_{solv}^{reorg}/k_B &= -4\pi\rho_w\left((\rho_c + \rho_w)\bar{V}_w - 1\right) \int (g_{nw} - 1) r^2 dr \\ &- 4\pi\rho_c\left((\rho_c + \rho_w)\bar{V}_c - 1\right) \int (g_{nc} - 1) r^2 dr - 4\pi\rho_w \\ &\times \int (g_{nw} \ln g_{nw}) r^2 dr - 4\pi\rho_c \int (g_{nc} \ln g_{nc}) r^2 dr\end{aligned} \tag{12}$$

Recent investigations have indicated facet-dependent ion and water adsorption on the bare gold nanoparticle in electrolyte solutions.[43,44] Such an observation was found for the polarizable force-field used for modeling gold interactions. However, in our work, where we have used non-polarizable force-field to model gold interaction, we find that the (100) facet is more hydrated and shows slightly higher ion adsorption than (111) facet due to the covalently bonded ss-DNA on (111) facet that makes it less accessible to water (and ions) (see Fig. S6).

A. Observables computed

Effective radius is determined by the average distance of each phosphorus (P) atom in the ss-DNA from the center of mass of the gold core, providing a measure of the overall size and shape of the system.

$$\langle R_{effect} \rangle = \frac{1}{N_{frames}} \sum_{N_{frames}} \sqrt{\frac{1}{N_P + 1} \left\langle \sum_{P=0}^{N_P} (\mathbf{r}_P - \mathbf{r}_G)^2 \right\rangle} \tag{13}$$

where, N_P represents the number of P atoms, r_P and r_G are the coordinates of the P and Au core center of mass respectively and N_{frames} denotes the number of frames in the trajectory.

Orientation angles Two angles, θ and ϕ are computed. To determine the orientation of the alkyl linker to the gold surface, angle formed between the gold surface and the thiol linker is defined by θ, where $\vec{a}$ is the position vector defined for the Au atoms on (111) facet functionalized with ss-DNA and $\vec{b}$ is the position vector defined by positions of the C bonded to SH group in the linker and the O bonded to P of the first thymine (next to the alkyl linker) as shown in Fig. 1.

$$\theta = \cos^{-1}\left(\frac{\vec{a}.\vec{b}}{|\vec{a}| * |\vec{b}|}\right) \tag{14}$$

To determine the orientation of the DNA chain relative to the thiol linker, ϕ is defined.

$$\phi = \cos^{-1}\left(\frac{\vec{b}.\vec{c}}{|\vec{b}| * |\vec{c}|}\right) \tag{15}$$

where $\vec{b}$ is position vector as defined before and $\vec{c}$ is position vector for the O bonded to P of the first thymine (next to the alkyl linker) and P of the second thymine residue (see Fig. 4). They can be used to gain insights into bending of the linker group and ss-DNA chain on the AuNP surface. If θ is small, it may indicate that the thiol linker is strongly bound to the gold surface, which could contribute to the stability of the system. If the angle ϕ is large, it may indicate that the DNA chain is not well-aligned with the thiol linker, which could affect its reactivity and binding properties. *Ligand shell entropy* is computed using Schlitter's method of using covariance in atomic positions of all the atoms of ss-DNA chains. The definition of this entropy, S_L, is[45]

$$S_L = \frac{1}{2} k_B \ln \det\left[\mathbf{1} + \frac{k_B T e^2}{\hbar} \mathbf{M}^{1/2} \sigma \mathbf{M}^{1/2}\right] \tag{16}$$

where, e is Euler number equal to exp(1), $\mathbf{M}$ is 3N-dimensional diagonal matrix having masses of N atoms of ss-DNA chains, T is the temperature of simulation, σ is the matrix with covariance in atomic positions of ss-DNA atoms. The GROMACS utility of *gmx covar* and *gmx anaeig* has been used to compute this entropy.

Solvent reorganization entropy is computed using the Kirkwood–Buff integrals and radial distribution functions (rdfs) for pair correlations between nanoparticle–water, nanoparticle–crowder, crowder–crowder, water–water, crowder–water are denoted by g_{nw}, g_{nc}, g_{cc}, g_{ww}, g_{cw} respectively as described in Eq. (12). For g_{nw} and g_{nc}, the correlations are computed between the center of mass of Au core and center of mass of water and the crowder. Similarly, center of mass of water or crowder are used for computing the g_{cc}, g_{ww}, g_{cw}. The rdfs are shown in Fig. 2. The rdf correction as given by Ganguly *et al.*[46] has been used to correct the g_{cc} since the KBI (G_{cc}) computed from g_{cc} was found to converge poorly. The corrected rdf is given by,[46,47]

$$g_{cc}^{corr}(r) = g_{cc}(r)\frac{N_j f(r)}{N_j f(r) - \Delta N_{ij}(r) - \delta_{ij}} \tag{17}$$

where, $f(r) = 1 - \frac{4}{3}\frac{\pi r^3}{V_{tot}}$ with $\Delta N_{ij} = \rho_j G_{ij}$ is the excess number of j particles within a radius r of i particle. The ρ_j is the number density of jth particle, G_{ij} is the KBI for correlation between i and j particles, V_{tot} is the total volume of the solution and δ_{ij} is Kronecker

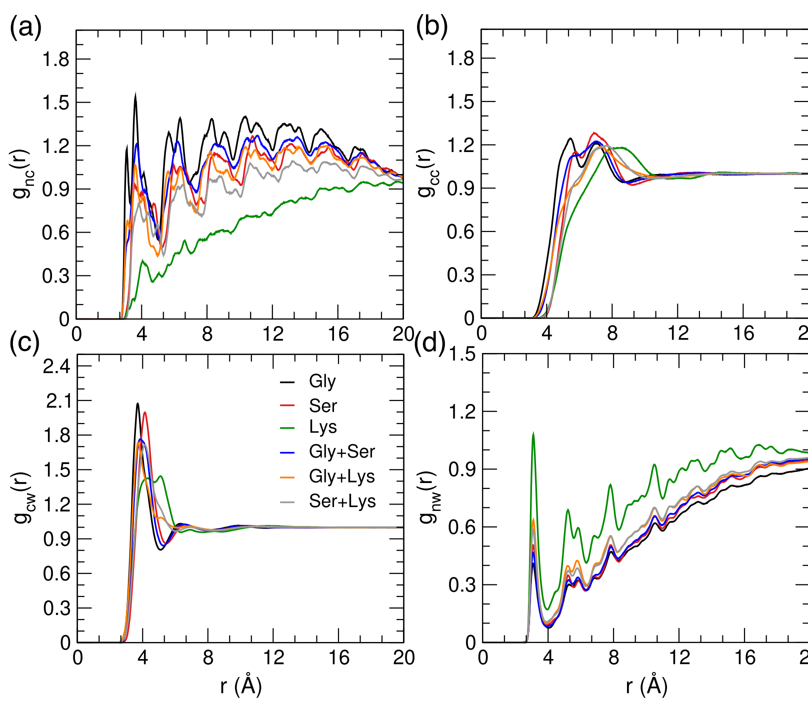

FIG. 2. Radial distribution functions of (a) nanoparticle (com of Au core)-crowder, (b) crowder–crowder, (c) crowder–water, (d) nanoparticle–water in crowded solutions of Gly as black, Ser as red, Lys as green and mixture of Gly+Ser as blue, Gly+Lys as orange, Ser+Lys as grey.

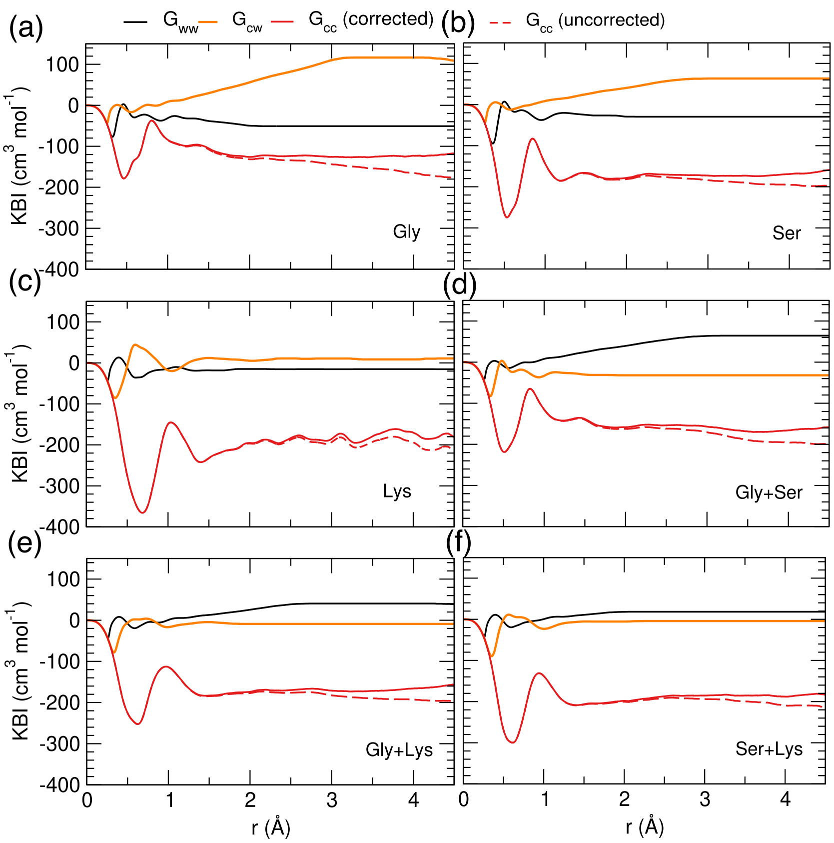

FIG. 3. Kirkwood–Buff integrals for crowder–crowder (G_{cc}), crowder–water (G_{cw}) and water–water (G_{ww}) interactions. The dotted lines show the uncorrected KBI Gcc and the solid line for G_{cc} represent the KBI after applying rdf correction.

delta. The KBIs computed for calculating the solvent reorganization entropy are shown in Fig. 3. The KBI for crowder–crowder interaction (G_{cc}) is seen to converge poorly. The rdf correction is seen to improve the convergence in G_{cc}. Using these KBIs, the partial molar volumes of crowder and water are computed that are used in Eq. (12) to compute the solvent reorganization entropy (ΔS_{solv}^{reorg}). Summing the ΔU_{solv} as computed from the trajectories and $-T\Delta S_{solv}$ provides the estimation of ΔG_{solv}.

IV. RESULTS AND DISCUSSION

The effective radius (R_{effect}) of the nanoparticle is found to be the least in pure water as shown in Fig. 4(a). On addition of single crowders, the R_{effect} increases in the order of increasing size of the crowder i.e. Gly < Ser ≈ Lys (also see Fig. S2). In the mixture of crowders, the R_{effect} increases in Gly+Ser mixture indicating an *additive* crowding effect. However, a *non-additive* effect is observed in Gly+Lys and Ser+Lys mixtures, where the R_{effect} decreases in Gly+Lys and does not change significantly in Ser+Lys mixture as compared to that in single crowder solutions. An increase in the packing fraction of Gly and Ser in the single crowder solutions to match the packing fraction of 34% in Lys solution, leads to an increase in the R_{effect} in Gly solution (see Fig. S8). The R_{effect} in Ser solution does not change significantly. The qualitative trend in R_{effect} between the crowder solutions remains the same i.e. with increasing crowder size, the R_{effect} increases. The solvation of the AuNP is directly correlated with the size of the AuNP where larger effective radius in AuNP supports favourable solvation in crowded solution as compared to that in pure water. This is indicated by the decreasing free energy of solvation (ΔG_{solv}) in Fig. 4(b). The solvation entropy (ΔS_{solv}) supports the ΔG_{solv} with $-T\Delta S_{solv} < 0$ for all the solutions. Relative to that in pure water, however, a loss in ΔS_{solv} (or an increase in $-T\Delta S_{solv}$) is observed in the Gly solution. This is overcompensated by the gain in the solvation energy (decrease in ΔU_{solv}). The ΔS_{solv} increases further for Ser and Lys solutions supporting ΔG_{solv} which is further supported by the decrease in ΔU_{solv} relative to that in pure water. However, in comparison to the Gly solution, the ΔU_{solv} becomes less favourable in Ser and Lys solutions. In the case of mixed crowded solutions, the ΔS_{solv} increases (or $-T\Delta S_{solv}$ decreases) and ΔU_{solv} increases in the order Gly+Ser < Gly+Lys < Ser+Lys. This indicates that the solvation in the mixed crowded solutions is entropically favourable and is opposed by the energy of solvation as the crowder size increases in the mixtures.

A further examination of the energetic contributions [Fig. 4(c)] shows that the adsorption of the crowders on the Au surface or

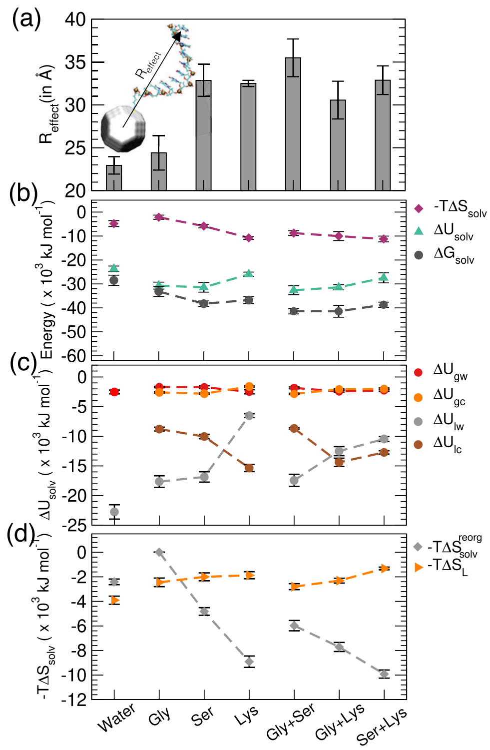

FIG. 4. (a) Effective radius of ss-DNA AuNP, (b) change in solvation free energy, energy and entropy, (c) components of the solvation energy change and (d) components of solvation entropy change in different crowder solutions.

ligand chains results in favourable gold–crowder energy (ΔU_{gc} < 0) and ligand–crowder energy (ΔU_{lc} < 0). Interestingly, the ΔU_{lc} is seen to decrease in single crowded solutions in the order Gly < Ser < Lys indicating that as the crowder size increases, the ligand-crowder interaction becomes more favourable and water is displaced from the ligand shells such that the ΔU_{lw} increases. In case of Lys solution the ΔU_{gc} increases, indicating less favourable Au–Lys interactions. In case of mixed crowded solutions, the ΔU_{lw} increases in the order Gly+Ser < Gly+Lys < Ser+Lys indicating that the bulkier crowders displace more water from ligand shells and result in higher loss of water–ligand contacts. The ΔU_{lc} is found to decrease for Gly+Ser and Gly+Lys mixtures, but an increase is observed for Ser+Lys mixture. Therefore, in the crowded solutions the increase in ΔU_{solv} observed in Fig. 4(b) is due to the increasing ΔU_{lw} that opposes the solvation free energy.

The favourable crowder–ligand interactions result in decreased conformational flexibility of the ligands leading to a decrease in the conformational entropy (ΔS_L) or an increase in $-T\Delta S_L$ [Fig. 4(d)] relative to that in pure water. The solvent reorganization entropy (ΔS^{reorg}_{solv}) is observed to decrease (or $-T\Delta S^{reorg}_{solv}$ increases) in Gly solution relative to that in pure water but increases ($-T\Delta S^{reorg}_{solv}$ decreases) in the Ser and Lys crowded solutions. In the mixed crowded solutions, the ΔS^{reorg}_{solv} increases relative to that in pure water in the order Gly+Ser < Gly+Lys < Ser+Lys. The presence of ions in solution is expected to play a role in the solvation thermodynamics. The nanoparticle–ion interactions have been accounted for in the ΔU_{solv}. However, we have not included the ion effects in the ΔS_{solv} term, since thermodynamic treatment to compute ΔS_{solv} for a four-component system is quite complex. Intuitively, it can be seen from Fig. S7, that the lower first peak height in the rdfs of gold–ion interaction indicates that there is less probability of ions to adsorb on the surface of gold in all single crowder solutions. The probability to find ions around the gold nanoparticle follows the order in the crowded solutions of Lys < Ser ≈ Gly. This indicates that the contribution of the gold–ion interaction will be small to the ΔS_{solv} and therefore can be neglected. This also may be because we are using non-polarizable gold force-field.

To further understand the trends in ΔS^{reorg}_{solv} and the effect on R_{effect}, we examine the rdfs of the crowder and ligands. An interesting differential local ordering of crowders is observed on the Au surface in single crowder solutions, although all crowders favourably interact with the ligand chains as well (Figs. S3 and S4). Lys being positively charged crowder tends to interact more with the negatively charged ss-DNA backbone. On the other hand, Ser has higher tendency to interact with the ss-DNA bases (Figs. S3 and S4). Figure 5(a) shows that the Gly has higher tendency to adsorb on the Au surface followed by Ser and Lys having the least probability due to the increasing crowder size. This results in a loss of entropic degrees of freedom of the crowders in the order Gly > Ser > Lys as reflected in the increasing ΔS^{reorg}_{solv} in the same order for the single crowder solutions. Such a higher ordering of Gly on Au surface results in Gly acting as a "glue" that favourably interacts with both the Au surface and the ligand chains leading to bending of the linker chains of the ss-DNA toward the Au surface, resulting in a smaller R_{effect}. Figure 5(b) shows that in Gly solution, there is higher probability of linker of chains exhibiting $\theta < 40^\circ$ and $\theta \approx 120^\circ$ which correspond to the *bent* conformations (observed mostly in pure water, Fig. S5). The ss-DNA chains also occupy $\phi \approx 80^\circ$ in case of Gly solutions. Since in the crowded solutions of Ser and Lys, there is lower crowder occupancy on Au surface, the "glue" effect of the crowder is not dominant. There is low probability of attaining $\theta < 40^\circ$ (bent conformation) in Ser solution as seen in Fig. 5(c). The probability of the chains attaining $\theta > 60^\circ$ and $\phi > 100^\circ$ is higher indicating *stretched* conformation of the chains. In the crowded Lys solution, a very strong probability is obtained for chains attaining $\theta > 60^\circ$ and $\phi > 100^\circ$, indicating many stretched ligand conformations. Although, a peak is observed for the bent linker conformation with $\theta > 40^\circ$ as well. The Ser and Lys adsorption on Au surface, however low, exerts steric effects and does not allow bending of linkers on Au surface.

Interestingly, in the mixed crowded solutions, differential ordering of crowders on Au surface results in additive and non-additive effects on R_{effect}. In solution of Gly+Ser, both the crowders have almost equal tendency to occupy the Au surface [Fig. 6(a)]. This is reflected in the less favourable ΔU_{lc} and more favourable ΔU_{gc} in Fig. 4(c) and lower ΔS^{reorg}_{solv} as compared to other mixtures [Fig. 4(d)], although Ser has a higher tendency to also interact with the ss-DNA chains (Figs. S3 and S4). This results in stretched ss-DNA conformations as can be seen in Fig. 6(d) that the chains do not acquire the bent conformation, but the linkers have $\theta > 60^\circ$ and 90° with $\phi > 100^\circ$ for all conformations resulting in a higher R_{effect}. This could be due to the steric effect of crowders on Au surface that do not allow the ss-DNA chains to bend. In Gly+Lys mixture, Gly has a

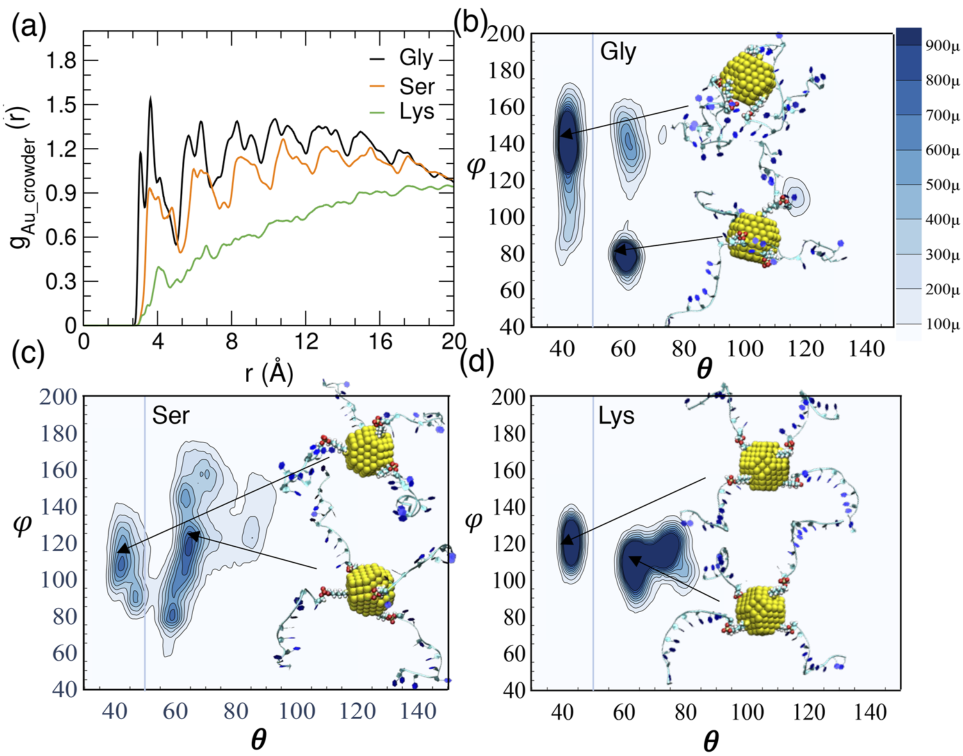

FIG. 5. (a) Radial distribution functions of crowders and Au surface in single crowder solutions and distributions of linker orientation angle relative to Au surface (θ) and ss-DNA orientation angle relative to the linker (ϕ) in crowded solutions of (b) Gly, (c) Ser and (d) Lys.

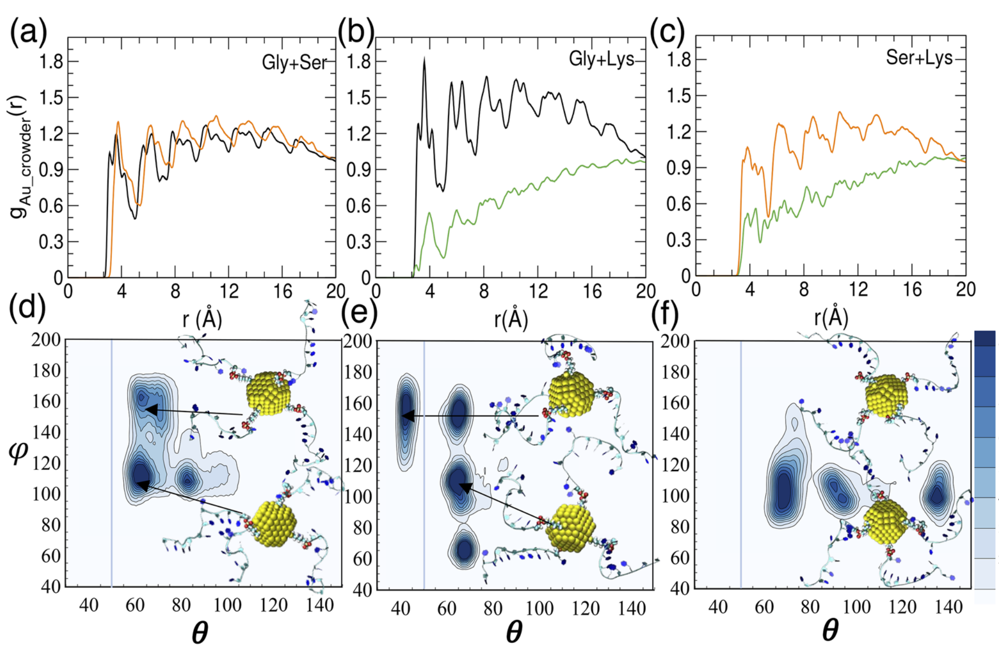

FIG. 6. Radial distribution functions of crowders and Au surface in (a) Gly+Ser, (b) Gly+Lys, (c) Ser+Lys. Distributions of linker orientation angle relative to Au surface (θ) and ss-DNA orientation angle relative to the linker (ϕ) in crowded mixtures of (d) Gly+Ser, (c) Gly+Lys and (d) Ser+Lys.

higher probability to adsorb on the Au surface [Fig. 6(b)] than Lys. This results in a higher ΔS_{solv}^{reorg} and favourable ΔU_{lc} (Fig. 4). However, in this mixture, bent conformations of the ligands are found with $\theta < 40^\circ$ as shown in Fig. 6(e). Additional bent conformations are observed with θ and ϕ around 70°. This results in a decrease in the R_{effect} in comparison to single crowder solutions, implying a non-additive crowding effect. The "glue effect" of Gly adsorbed on the Au surface dominates in this mixture, and bends the linkers toward the Au surface. An intriguing non-additive effect in R_{effect} is observed in Ser+Lys mixture. Ser has higher probability to adsorb on the Au surface than Lys [Fig. 6(c)]. Here, the R_{effect} does not significantly reduce than in the single crowder solutions, but also does not increase as much as in the case of Gly+Ser mixture. Bent linker conformations are observed with $\theta \approx 140^\circ$ as shown in Fig. 6(f). However, a significant number of conformations acquire $\theta \approx 60^\circ$ and 90° with $\phi > 90^\circ$ in this mixture. This could be due to the steric effects of large-sized Ser crowders on the Au surface that do not allow much bending of the ss-DNA chains.

V. CONCLUSIONS

Therefore, our results show that the crowder size and chemistry difference lead to a differential local ordering of crowders on the Au surface that leads to either bending or stretching of the ss-DNA chains. The small-sized crowder would have higher tendency to adsorb on Au surface (in single or mixed crowders) and would cause bending of the chains via cohesive interactions with the surface and ss-DNA chains. In the mixture of similar sized crowders (like Ser+Lys), where both the crowders have similar tendency to adsorb on Au surface, the steric effects of crowders would oppose bending of the ss-DNA chains. The solvation of nanoparticle becomes free energetically more favourable for larger effective size. The solvation process is entropically dominated, with the solvent reorganization entropy becoming more favourable with the increasing crowder size (and effective size of nanoparticle). An interplay of the packing fractions of the amino acids (crowders) and their size is expected to play a crucial role in determining the effective size of the functionalized gold nanoparticles. This would be an important controlling factor to design such nanoparticles for tailored biomedical applications. It would also be interesting to explore the temperature effects on the structure of the AuNP. Increasing the temperature would lower the preferential ordering of crowders on gold surface, especially in the solution of small-sized Gly crowders. This would be expected to reduce the "glue" effect of the crowders and would contribute to increase the effective size of the nanoparticle. On the other hand, an increase in the conformational entropy of the ligand chains may tend to collapse the chains on the gold surface resulting in a decrease in the effective size. Therefore, it would be interesting to validate the interplay of these two competing factors in determining the effective size of the nanoparticle at higher temperatures. Moreover, increasing the temperature would lead to a more favourable solvation entropy and more favourable solvation free energy would be expected for the nanoparticle.

Our results hold implications in designing self-assembled ss-DNA functionalized AuNPs with non-base pairing employing the mixed crowded environment as a tool. For this, there are two competing factors.[24] First, that the stretched ss-DNA chains result in larger effective AuNP radius and effective volume. This favours the AuNP self-assembly since that would result in an increase in the free volume in packed crowded solutions. Second, higher conformational flexibility of the ligand chains would attenuate (or oppose) the self-assembly of nanoparticles. Our results have shown that in a crowded solution, homogeneous or mixed, the conformational flexibility of ss-DNA chains is lowered, and the effective radius increases, enhancing the probability of ordering in nanoparticles in a multi-nanoparticle solution. Therefore, both these factors will contribute in driving the self-assembly of nanoparticles. The type and size of crowders in a mixed crowded environment can be used as a tuning factor to regulate the effective volume of the nanoparticle determining the extent of self-assembly of the nanoparticles.

SUPPLEMENTARY MATERIAL

The supplementary material includes snapshots of the crowded solutions, variation of effective radius of the nanoparticle with time, rdfs of crowders and ss-DNA chains and probability distribution of the orientation angles of ss-DNA chains in pure water solution, rdf of water and ions with (100) and (111) facets of Au nanoparticle, rdf of Au-ions, the effective radius of nanoparticle at same as high packing fractions of the system and rdf of ss-DNA chains with water.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Sibasankar Panigrahy: Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Writing – review & editing (equal). **Divya Nayar**: Conceptualization (lead); Funding acquisition (lead); Project administration (lead); Supervision (lead); Writing – original draft (lead).

DATA AVAILABILITY

The data that supports the findings of this study are available within the article and its supplementary material.

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